Understanding artifacts in MRI

Types of artifacts

Tissue-related Artifacts

Motion-related Artifacts

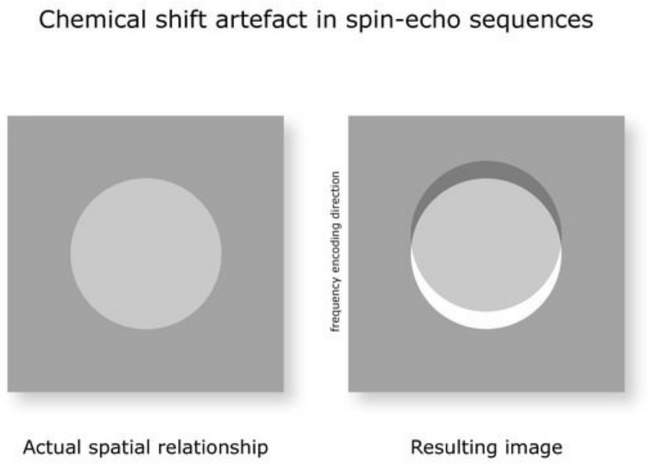
Technique-related artifacts

Ref: https://mriquestions.com/hellipmr-artifacts.html

Type	Artifact	Implementation method	Change in mask/ground truth Yes/No
Tissue-related	Chemical-shift artifacts Out of Phase Artifact		
Tissue-related	Susceptibility artifacts		
Tissue-related	Metallic artifacts		
Tissue-related	Dielectric artifacts		
Motion-related	Ghost Artifacts		

Type	Artifact	Implementation method	Change in mask/ground truth Yes/No
Technique-related	Partial Voluming Artifact		
Technique-related	Cross talk / Cross Excitation		
Technique-related	Shading artifact		
Technique-related	Truncation / Gibbs artifact		
Technique-related	Zipper Artifact		
Technique-related	Aliasing (wrap around)		
Miscellaneous	Central Point Artifact		
Miscellaneous	Data Clipping		
Miscellaneous	Data Error Artifacts		

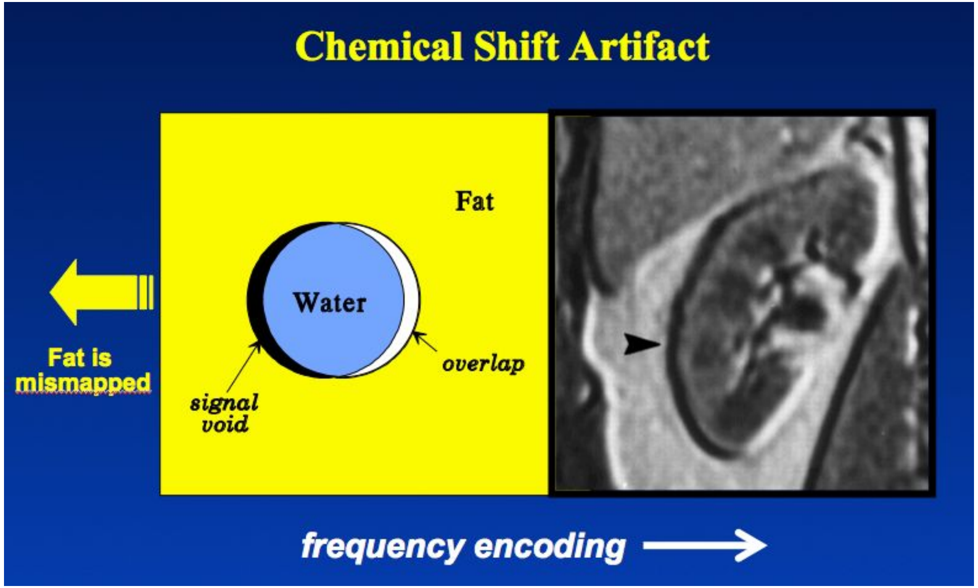
Tissue-related Artifacts :Chemical Shift Artifacts



Chemical shift is due to the differences between resonance frequencies of fat and water.

Ref: https://radiopaedia.org/articles/chemical-shift-artifact-1

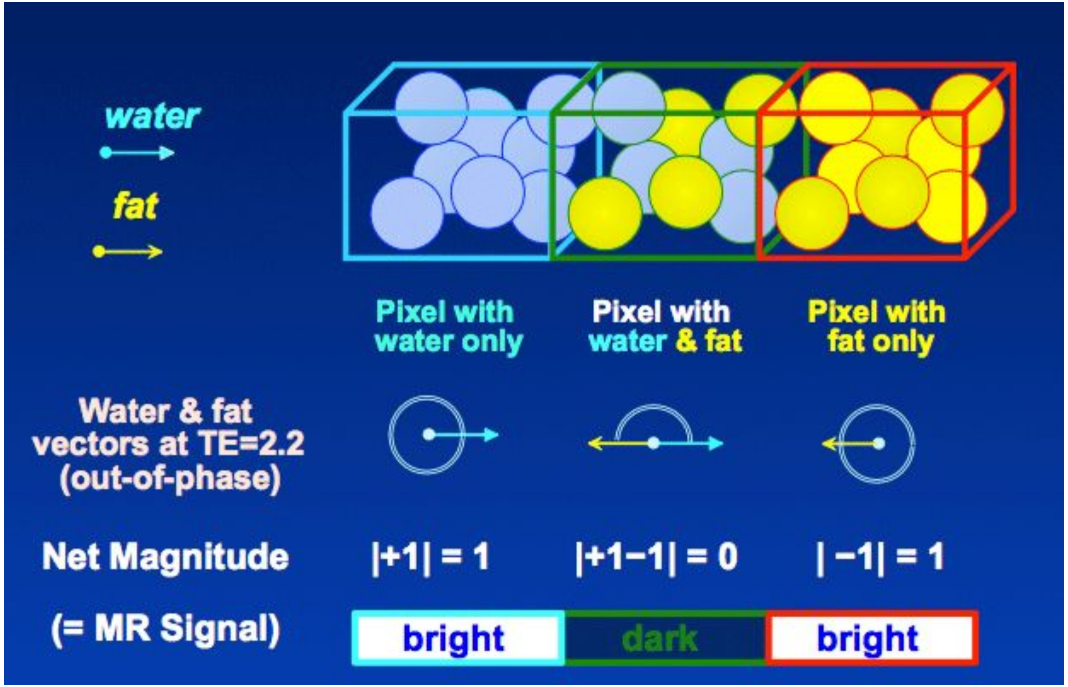
Tissue-related Artifacts : chemical shift artifacts



Ref: https://radiopaedia.org/articles/chemical-shift-artifact-1

Ref https://mriquestions.com/chemical-shift-artifact.html

Tissue-related Artifacts : Chemical Shift: 2nd Kind (Out of Phase Artifact)



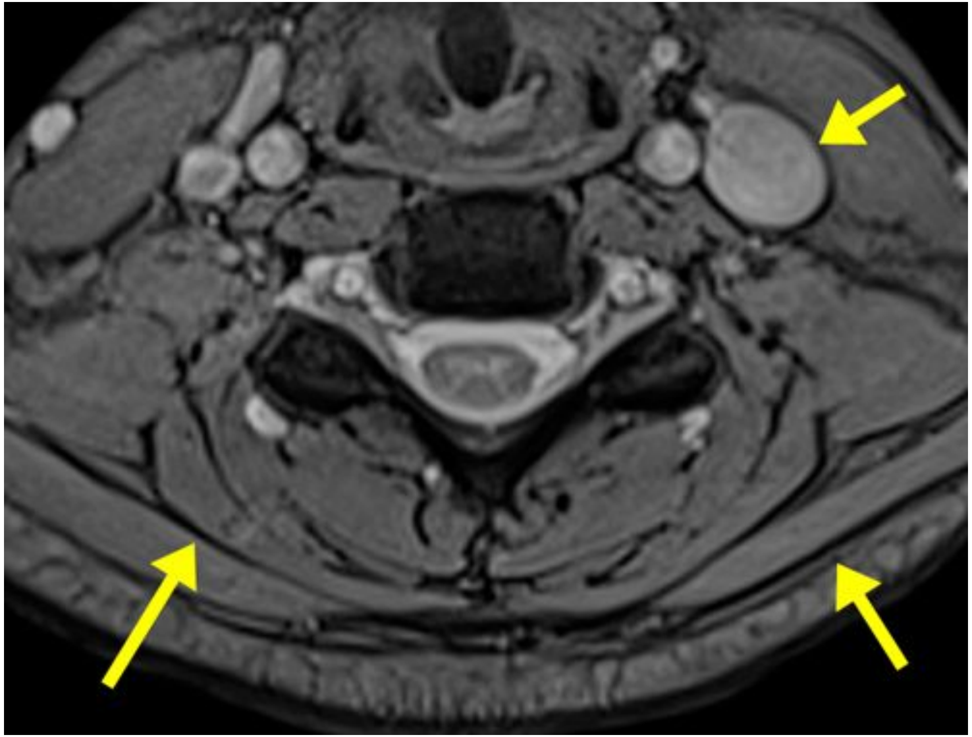
Because the water and fat signals are out of phase, they cancel each other producing zero net magnitude and hence a signal void.

Pure fat and pure water pixels have no such cancellation, and because magnitude image reconstruction is displayed, both are rendered bright in the final image.

Thus the pure water and fat pixels are separated by a dark boundary at their interface.

Ref: https://radiopaedia.org/articles/chemical-shift-artifact-1

Tissue-related Artifacts :Chemical Shift: 2nd Kind



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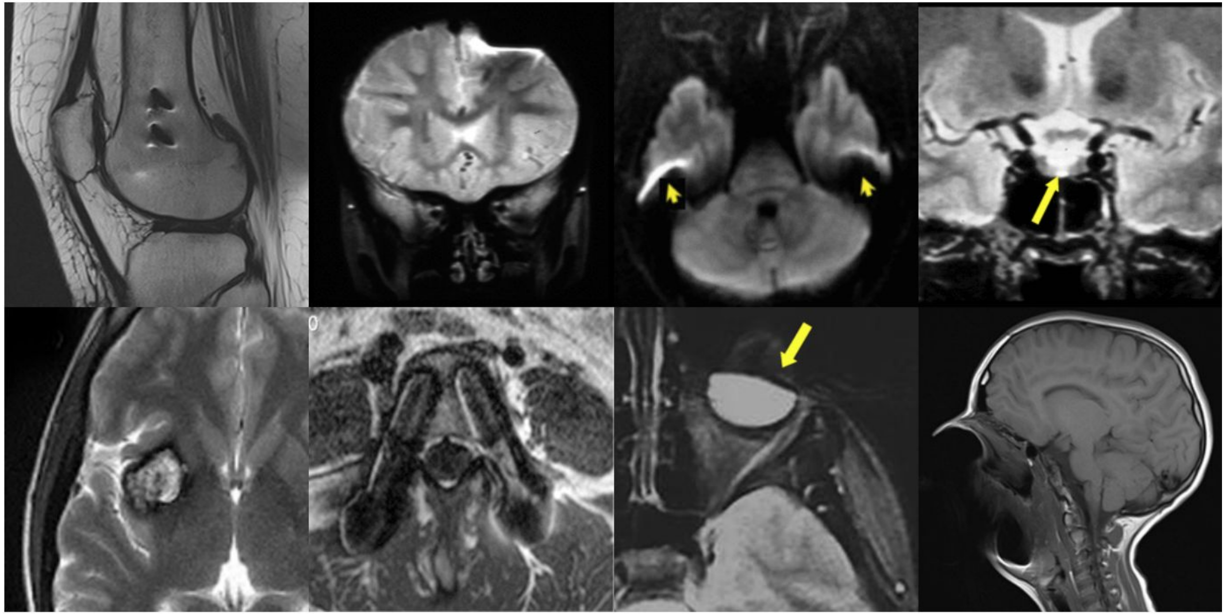
Ref: https://radiopaedia.org/articles/chemical-shift-artifact-1

Tissue-related Artifacts Susceptibility Artifact



- Artifacts due to differences in magnetic susceptibilities (χ) of tissues/materials
- Seen especially around metal implants and at air-tissue or air-bone interfaces
- Characterized by geometric distortion, very dark and very bright areas

The more severe metal artifacts may be reduced by specific metal artifact reduction pulse sequences

https://mriquestions.com/susceptibility-artifact.html

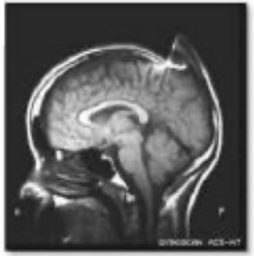
Tissue-related Artifacts : Metal Artifact Suppression

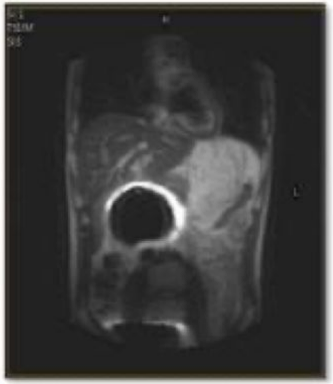
As the important first step in reducing metal-related artifacts, use the lowest field strength magnet possible

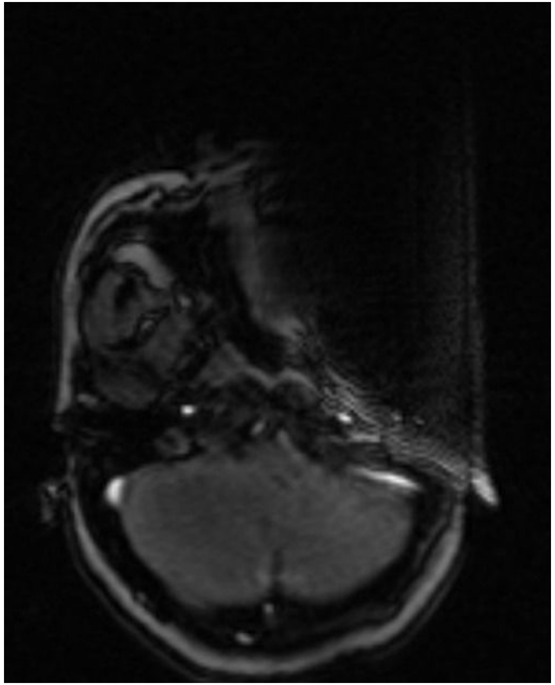
They are especially encountered while imaging near metallic orthopedic hardware or dental work, and result from local magnetic field inhomogeneities introduced by the metallic object into the otherwise homogeneous external magnetic field B_0.

These local magnetic field inhomogeneities are a property of the object being imaged, rather than of the MRI unit itself.

Examples of metallic artifacts origins





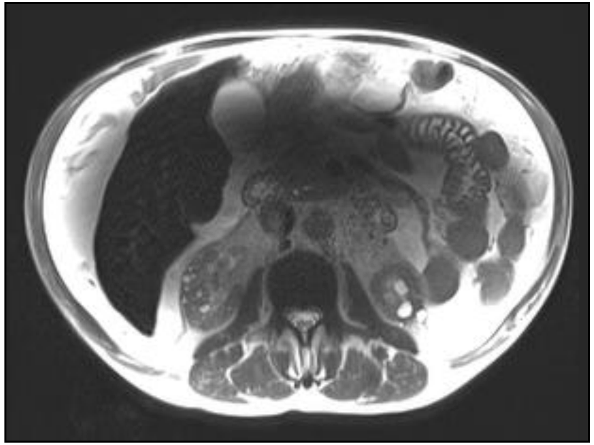


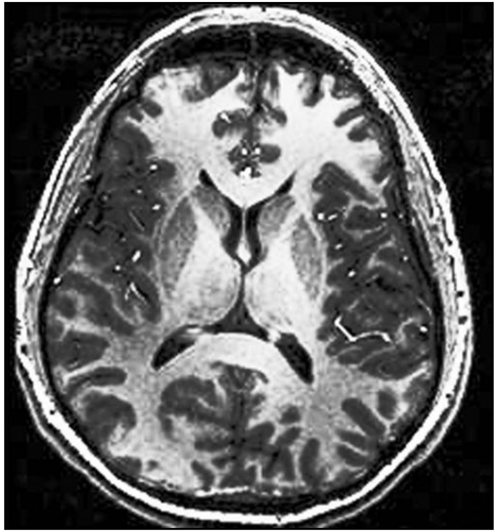
- Internal
  - Surgical clips, staples, shunts
  - Dental implants and appliances
- External
  - Jewelry, buckles
  - Piercings
  - Metallic stitching in clothing
  - Hairpins, some hairpieces, wigs
  - Make-up, tattoo

Warning: these items may also cause a burn risk

https://tsrt.or.tn/a/wp-content/uploads/2019/05/Lecture-Meet-the-Artifacts.pdf
https://radiopaedia.org/cases/27323/studies/27507?lang=us&source_of=https%3A%2F%2Fradiopaedia.org%2Farticles%2Fmagnetic-susceptibility-artifact

Tissue-related Artifacts :Dielectric Artifact



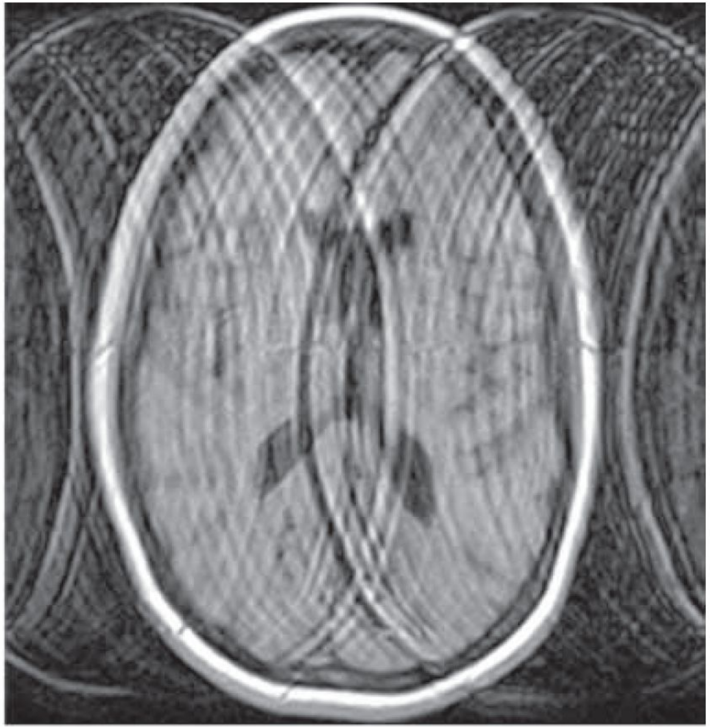


Abnormal bright and dark areas due to **B1** field inhomogeneity are frequently noted at very high fields (3T and above).

Dielectric pads are simple pouches containing a highly conductive material that are placed between the patient and the receiver coil to reduce dielectric artifacts.

Our modern 3T scanners employ dual RF technology, in which two separate sources with adjustable phase shifts are used transmit the RF-field. This technology has remarkably reduced image inhomogeneity effects

Motion-related Artifacts : Ghost Artifacts / Smearing



a significant ghosting artifact due to patient motion. Motion-related artifacts typically are propagated in the phase encoding direction

Pulsatile flow of blood or CSF, cardiac motion, and respiratory motion are the most important patient-related causes of ghost artifacts in clinical MR imaging.

Ref: An Image-based Approach to Understanding the Physics of MR Artifacts

cardiac and respiratory gating

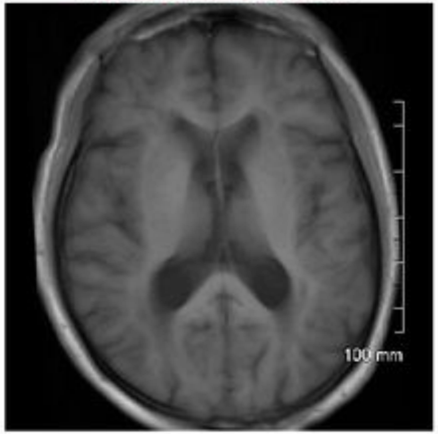
Technique-related Artifacts: Partial Voluming

Partial volume artifact occurs when many different tissues are averaged into one voxel. This may cause a misrepresentation of anatomy and image blurring.

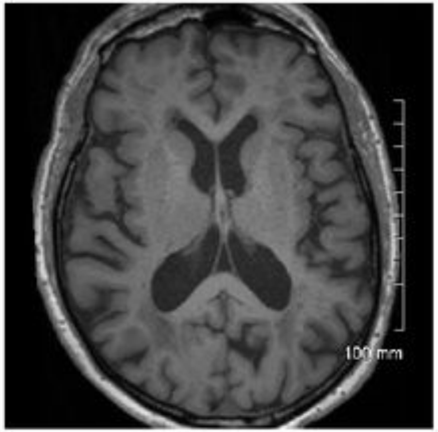
This artifact is dependent on the slice thickness chosen by the technologist. Thicker slices will introduce more opportunities for different types of tissues to be placed within a voxel.

20mm Slice Thickness

Partial Volume Artifact

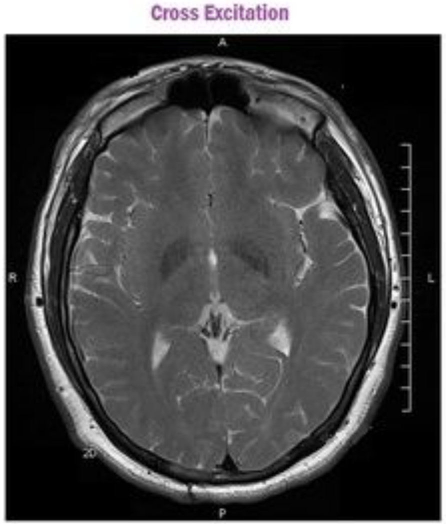


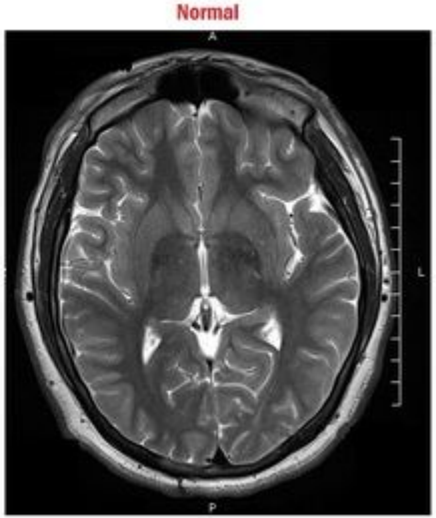
2mm Slice Thickness



Ref: http://www.mrishark.com/partial-volume.html

Technique-related Artifacts: Cross talk / Cross Excitation

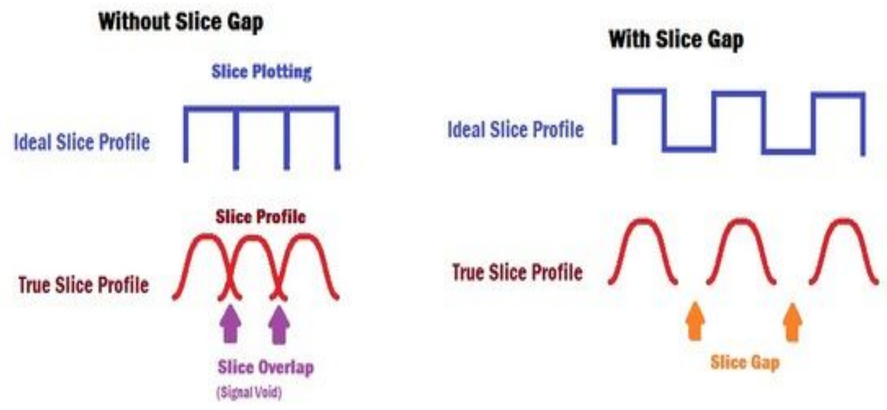




This can also be caused when slices are collected consecutively.

This means that part of one slice may overlap another slice. This will cause a loss of signal in our slice.

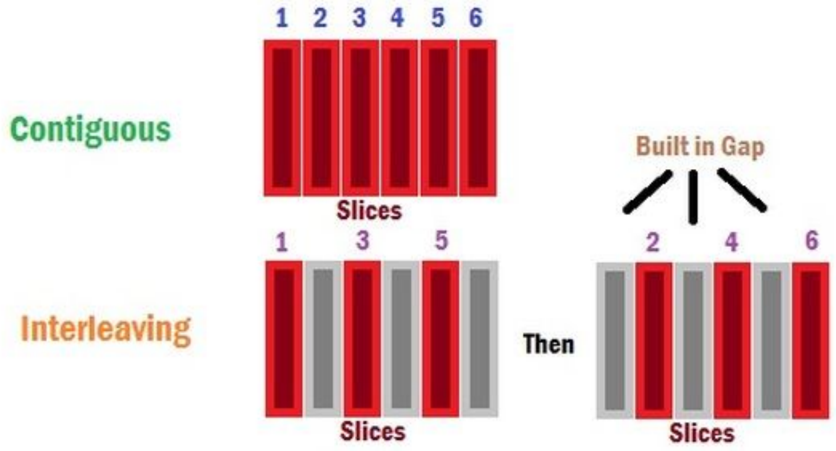
Ref http://www.mrishark.com/artifact.html



Introduce a slice gap

By using a slice gap of at least 30%, cross excitation can be reduced.

Using a slice gap can also cause other problems as well such as possibly missing pathology.

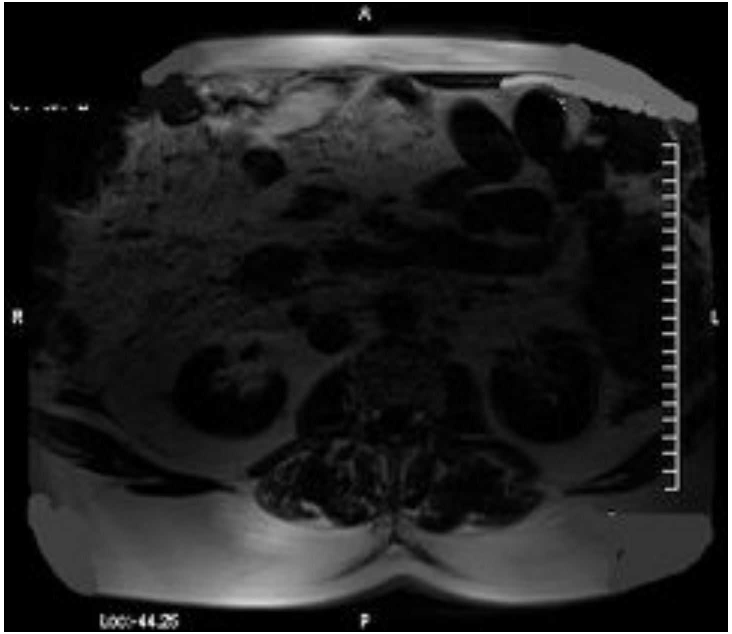


Interleaving slices

This will allow us to collect the odd number slices first and collect the even numbers second.

This technique will therefore build a slice gap into the data acquisition without needing to use a slice gap.

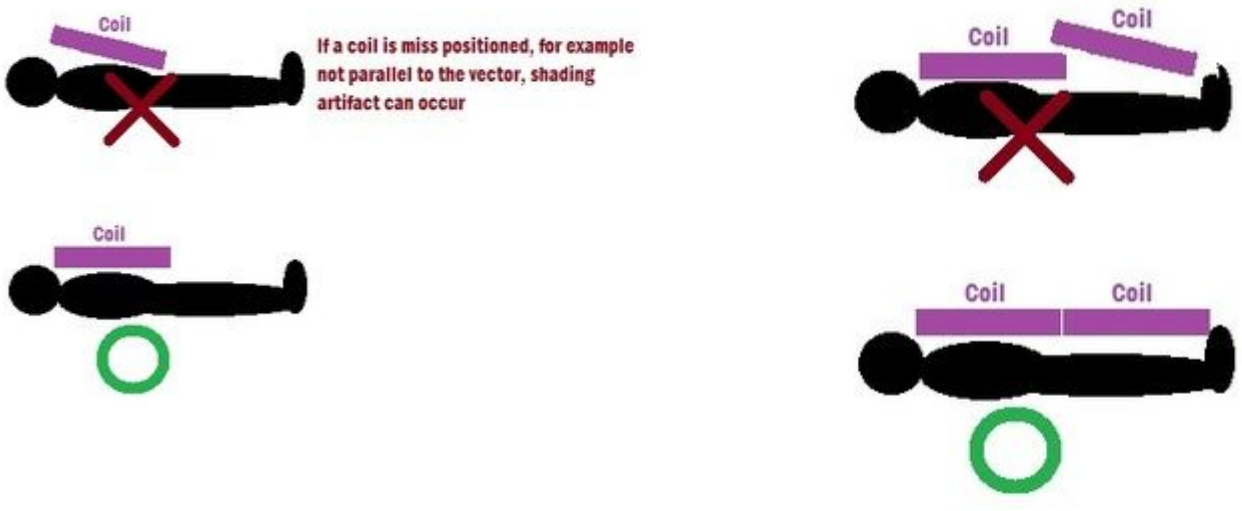
Technique-related : Shading artifact



This artifact is represented by different signal intensities that are found at different location in our image.

Patients who are very large can promote many of the issues associated with this artifact. This can be caused by a number of reasons.

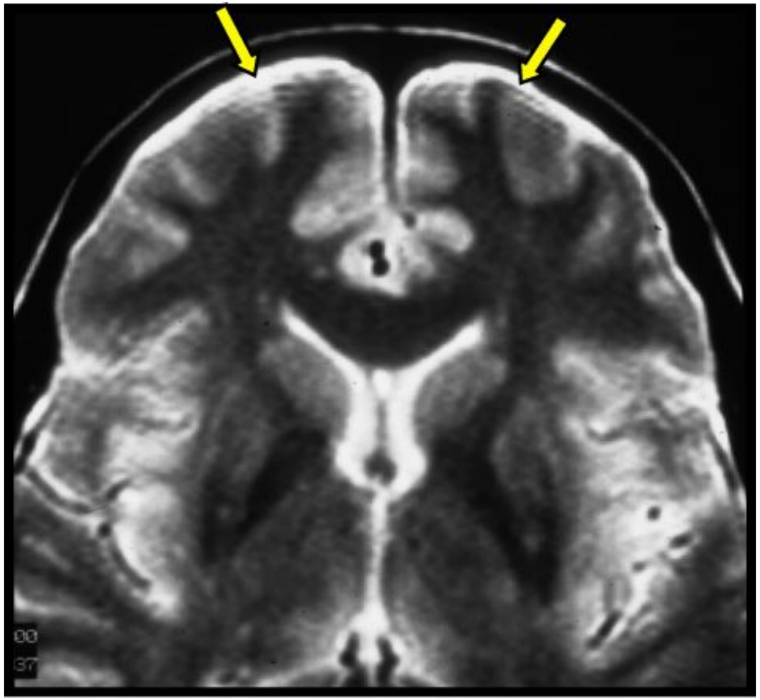
Technique-related: Shading artifact



Coil positioning

If a coil is not positioned properly, an uneven distribution of signal may be seen in our image. Also, if we are coupling coils improperly, signal may be lost at improper coupled coil point.

Technique-related: Truncation / Gibbs Artifact(***truncation***, ***ringing***, or ***spectral leakage*** artifacts)

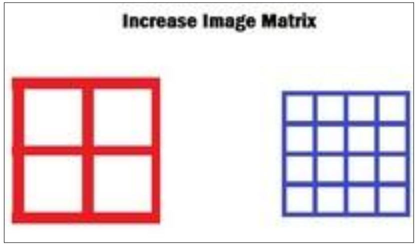


Truncation / Gibbs artifact is created when we undersample our image

Typically appear as multiple fine parallel lines immediately adjacent to high-contrast interfaces

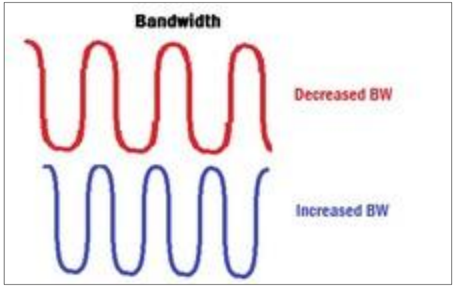
Ref : https://mriquestions.com/gibbs-artifact.html

Remedy for Gibbs Artifact



Increase the image matrix

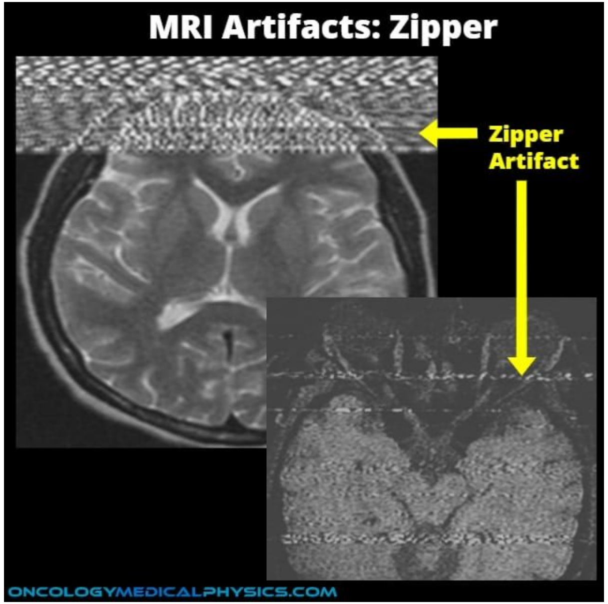
By increasing the amount of pixels in our image, we may better represent these abrupt changes. We should also try to create isotropic pixels in our image.



Increase Receiving Bandwidth

By increasing our receiving bandwidth we will collect more frequencies per pixel. (see chemical shift artifact for more information)

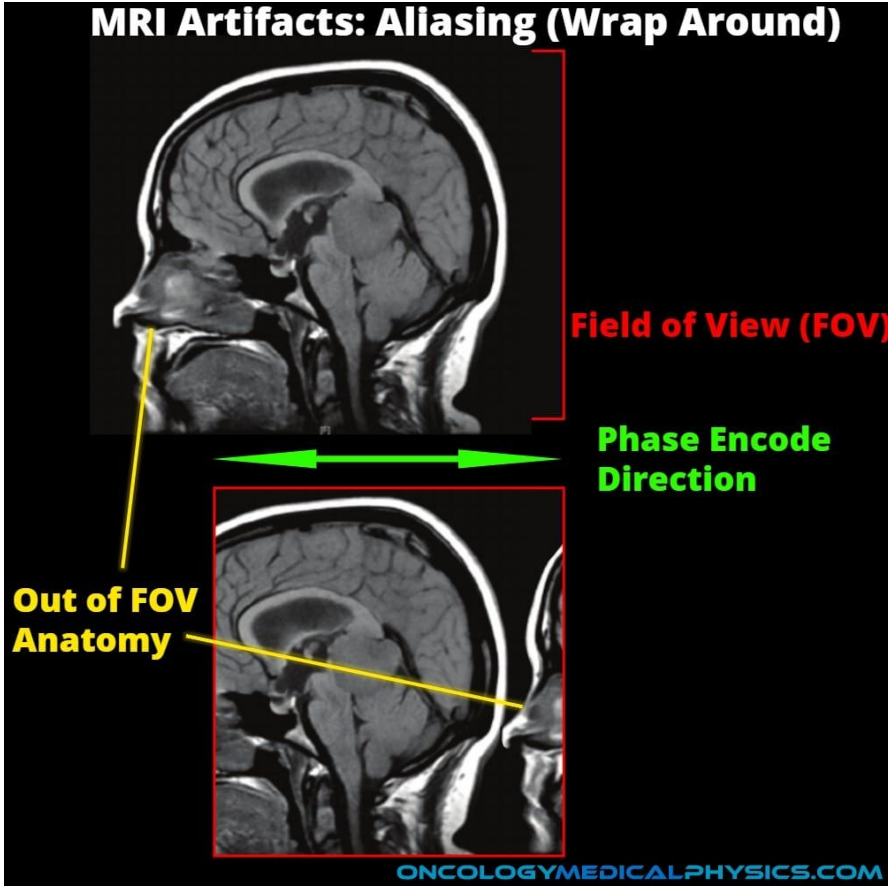
Technique-related : Zipper Artifact



If there is a leak or a breach of this faraday cage, radio frequences will enter the MRI suite and can contaminate the frequencies collected by our receiver coil.

Ref : https://oncologymedicalphysics.com/mri-artifacts/

Aliasing on MRI, (wrap-around)



Appearance

- Anatomy outside of the field-of-view (FOV) is superimposed on the opposite side of the image.

Causes

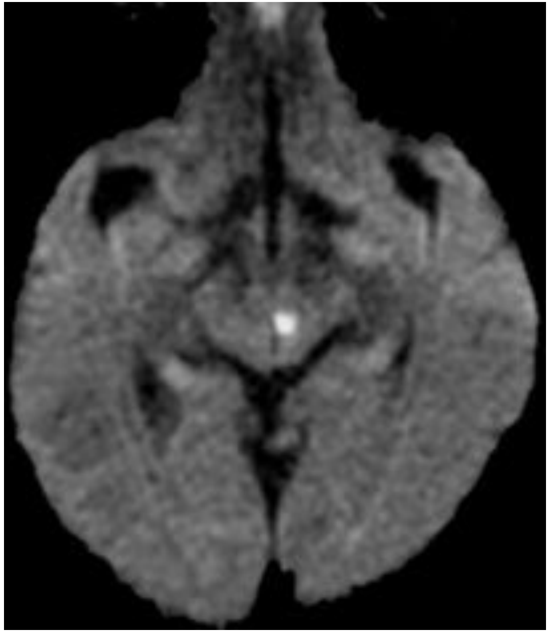
- Sampling at below the Nyquist frequency (< twice the RF frequency of the voxel) causes the apparent frequency to be lowered in a process known as aliasing.

Artifact Reduction

- Many scanners have automatic wrap around removal either by oversampling or by applying a low pass filter prior to analog-to-digital conversion.
- Change phase encode direction.
- Increase field-of-view.

Miscellaneous Artifacts

Central Point Artifact

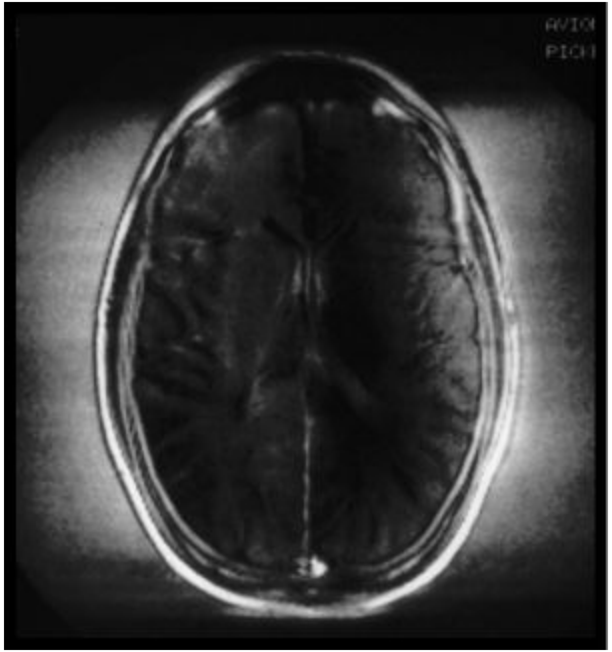


The ***central point artifact*** appears as either a bright or a dark dot precisely at the center of the image

It results from a constant direct current (DC)-offset in the receiver electronics that when Fourier transformed becomes a small spike in the center of the image.

Miscellaneous Artifacts

Data Clipping



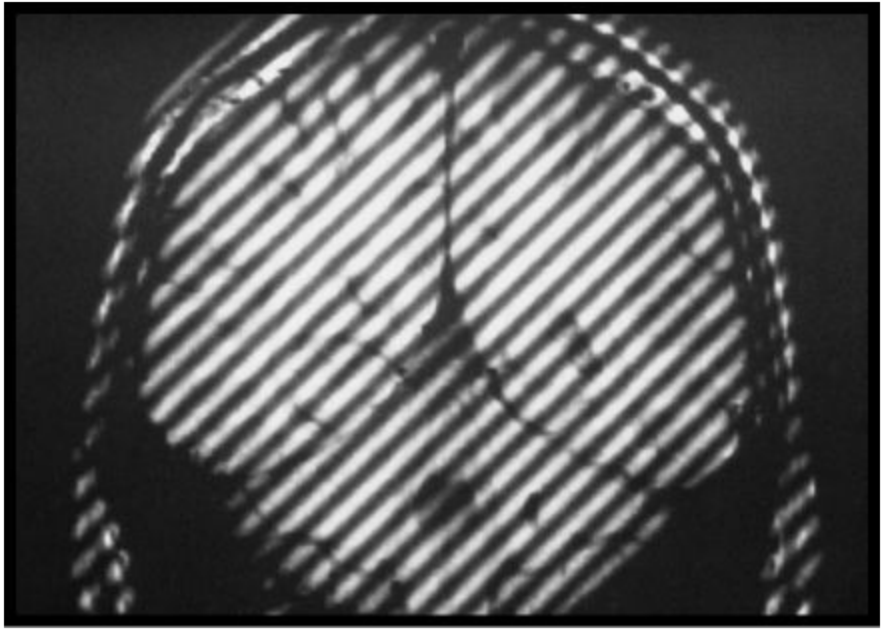
Improper calibration of the receiver amplifier gain can result in data clipping and an eerie, phantom-like image with a gray background.

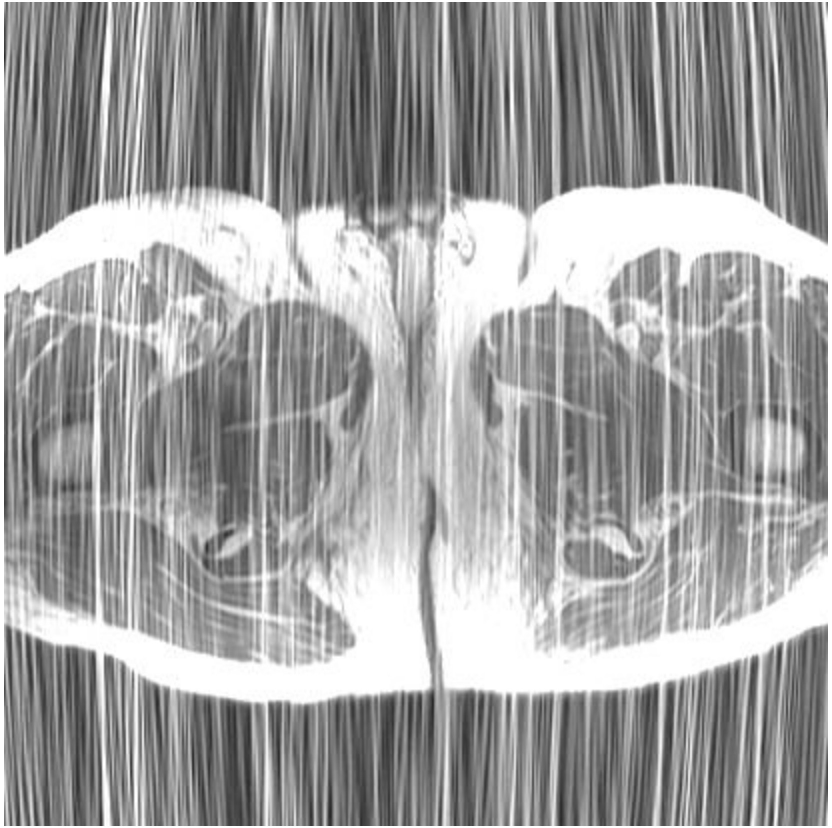
This artifact occurs whenever the gain setting is too high and the RF-signal exceeds the upper and lower cut-off levels of the RF amplifier ($\pm V_{max}$).

Careful recalibration and rescanning are the only means of overcoming this artifact.

Miscellaneous Artifacts

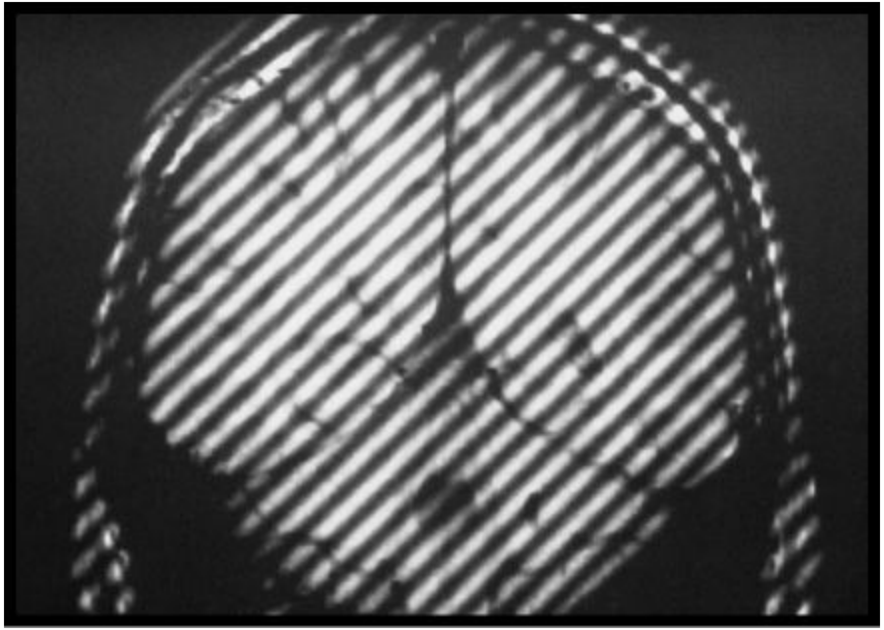
Data Error Artifacts





Miscellaneous Artifacts

Data Error Artifacts



From time-to-time a random "glitch" occurs in the data processing chain resulting in an image containing multiple lines that may be arrayed in a corduroy or herringbone pattern.

A common cause is ***spike noise*** in the raw data, whose Fourier transform is a series of stripes. Sometimes simply reprocessing the raw data will remove this artifact and salvage an otherwise unreadable group of images.

QC paper

```
The underlay/overlay pair of datasets (sub-030_T1w.nii.gz/sub-001_T1w.nii.gz) have oblique angle
difference of 3.099962 degrees. This may cause them to appear out of alignment
in the viewer.

 If you are performing spatial transformations on an oblique dset,
 or viewing/combining it with volumes of differing obliquity,
 you should consider running:
    3dWarp -deoblique
 on this and other oblique datasets in the same session.

** Other oblique warnings will be muted because AFNI_ONE_OBLIQUE_WARNING
   is set to YES. Consider setting it back to NO, obliquity warnings
   are much less overwhelming now.

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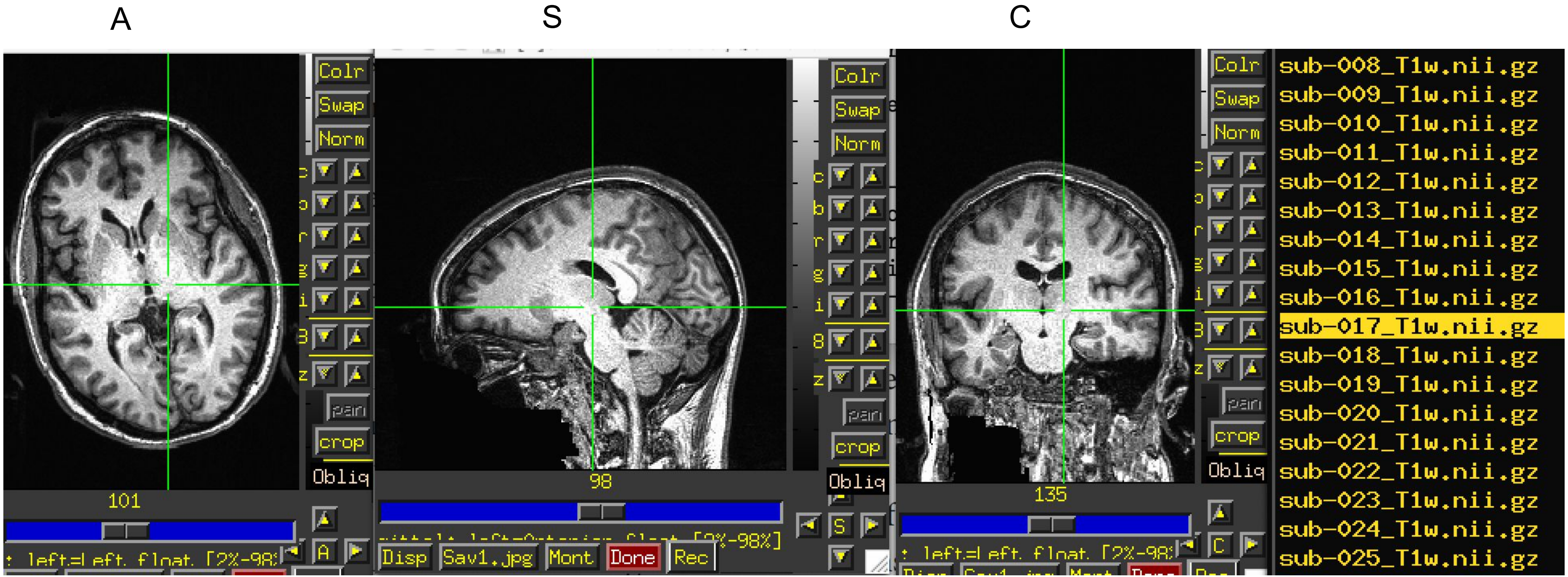
Overly tight FOV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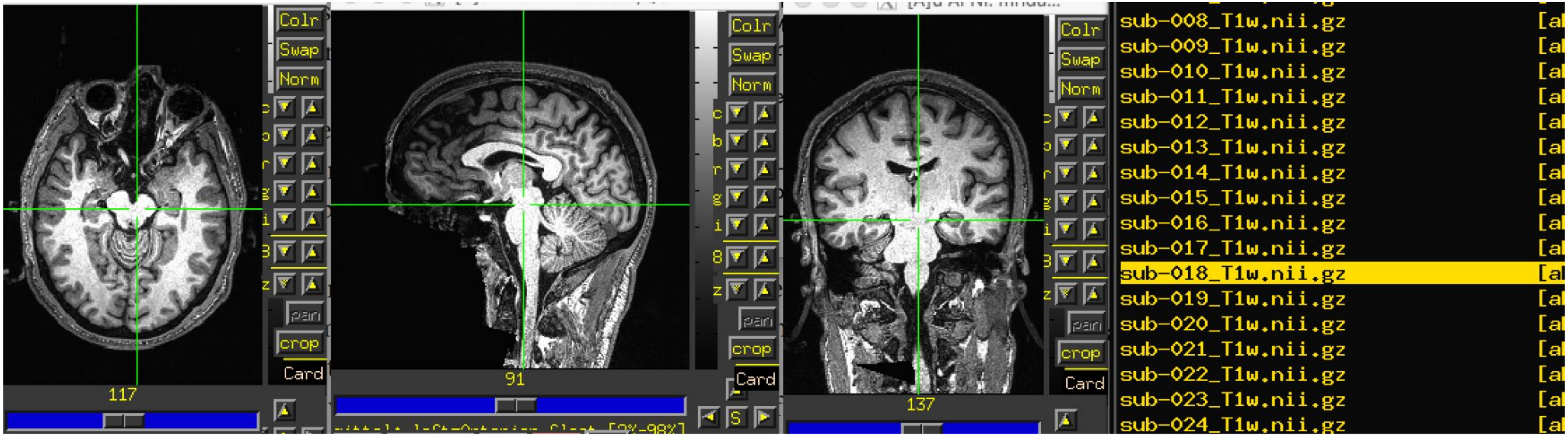
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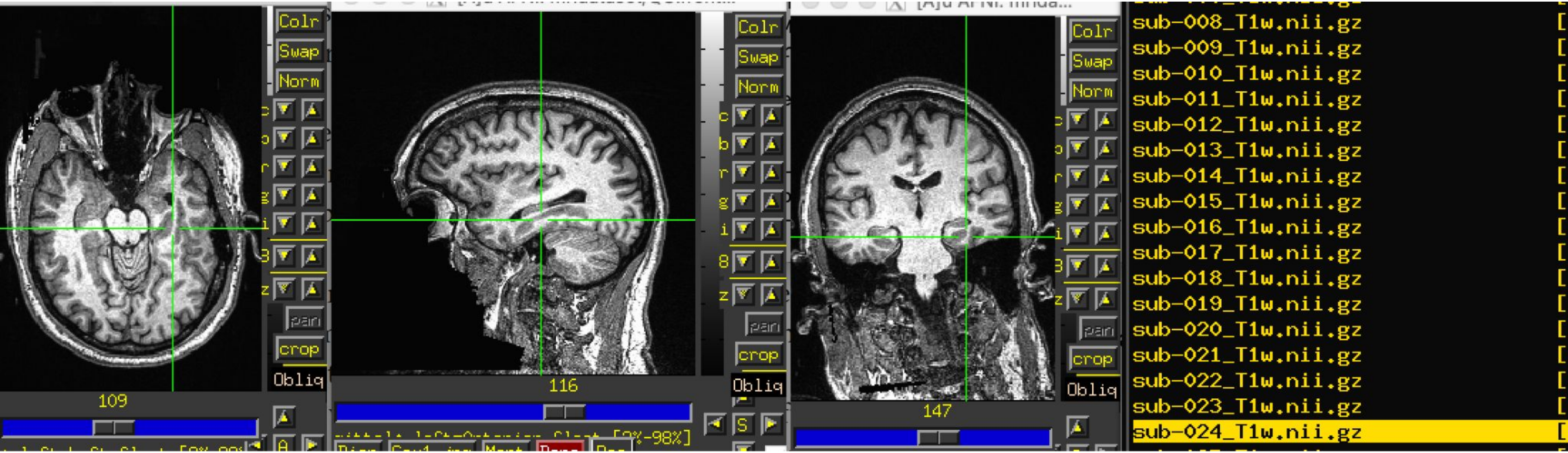
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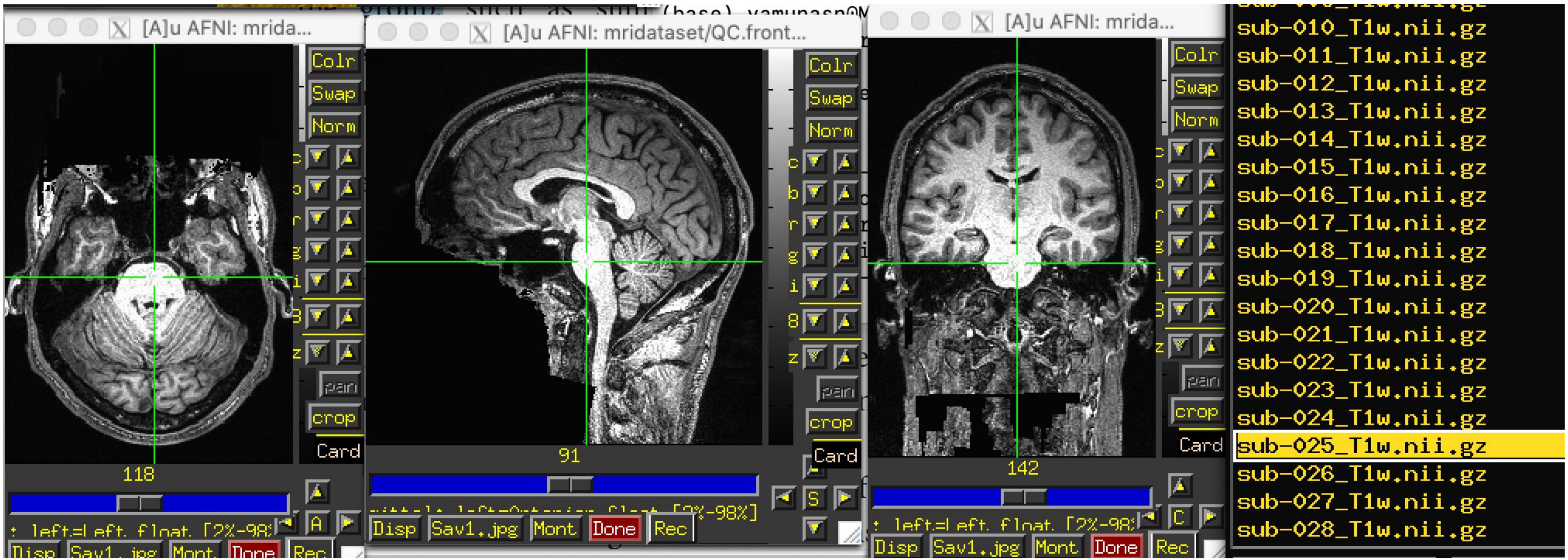
Oblique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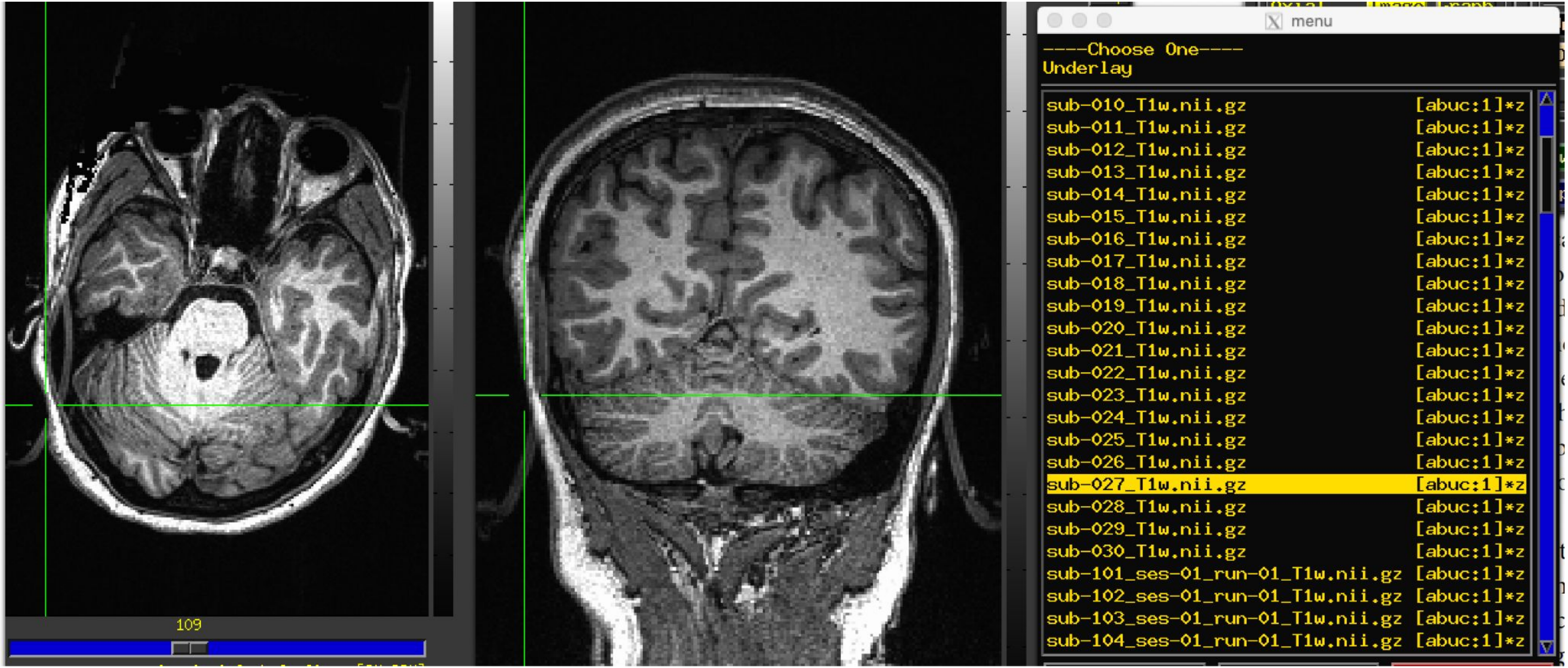
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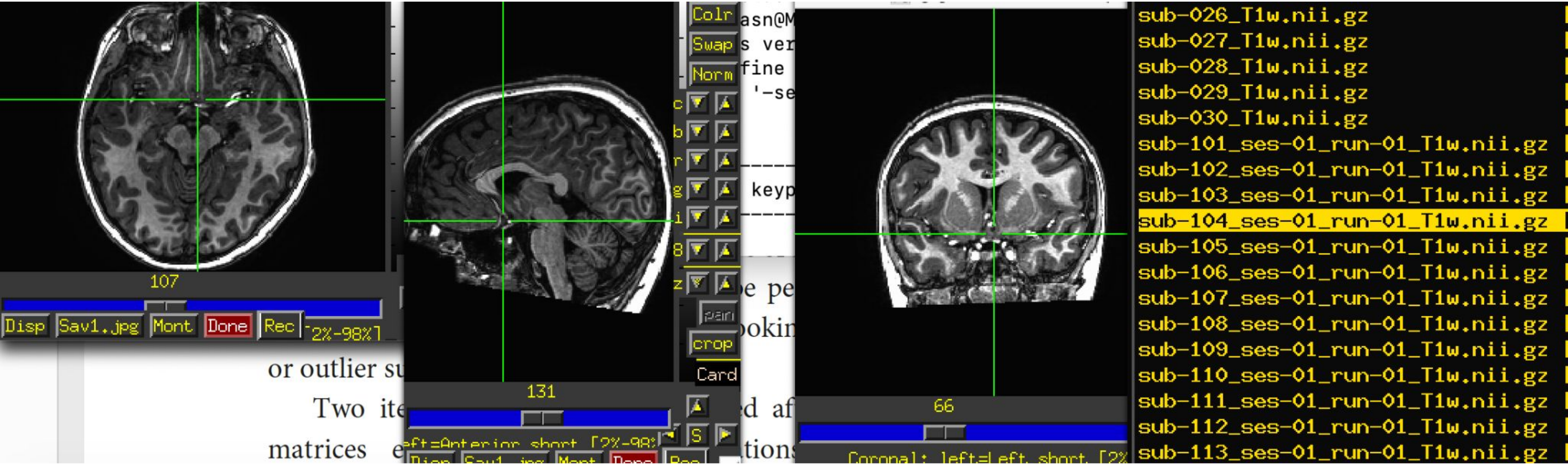
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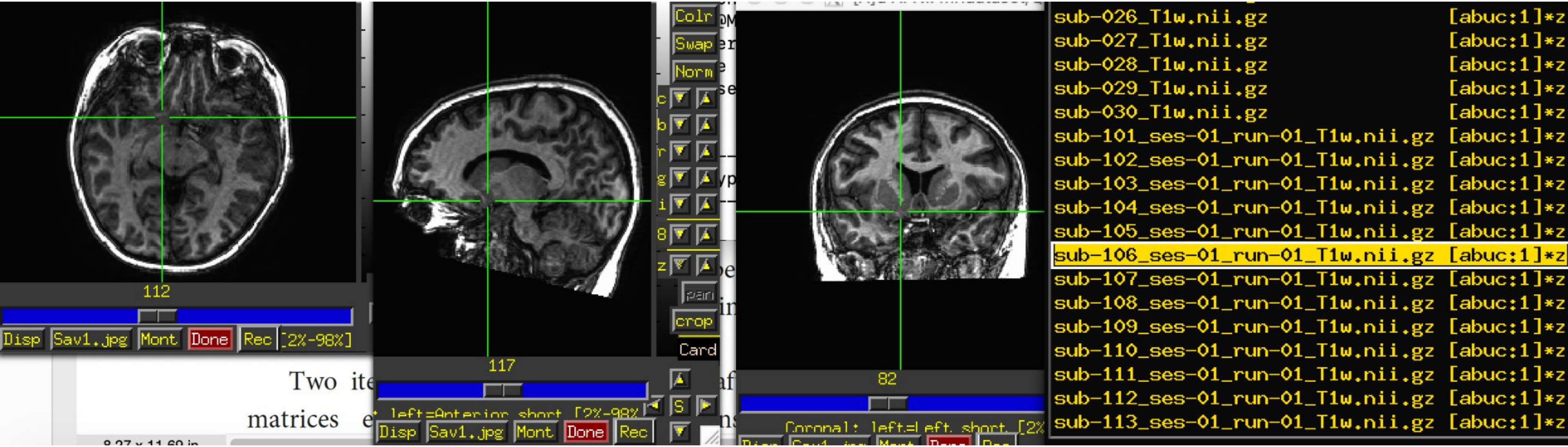
sub-110_ses-01_run-01_T1w.nii.gz
sub-111_ses-01_run-01_T1w.nii.gz
sub-112_ses-01_run-01_T1w.nii.gz
sub-113_ses-01_run-01_T1w.nii.gz


112
Disp
Sav1.jpg
Mont
Done
Rec
[2%-98%]
117
Colr
Swap
Norm
crop
Card
82
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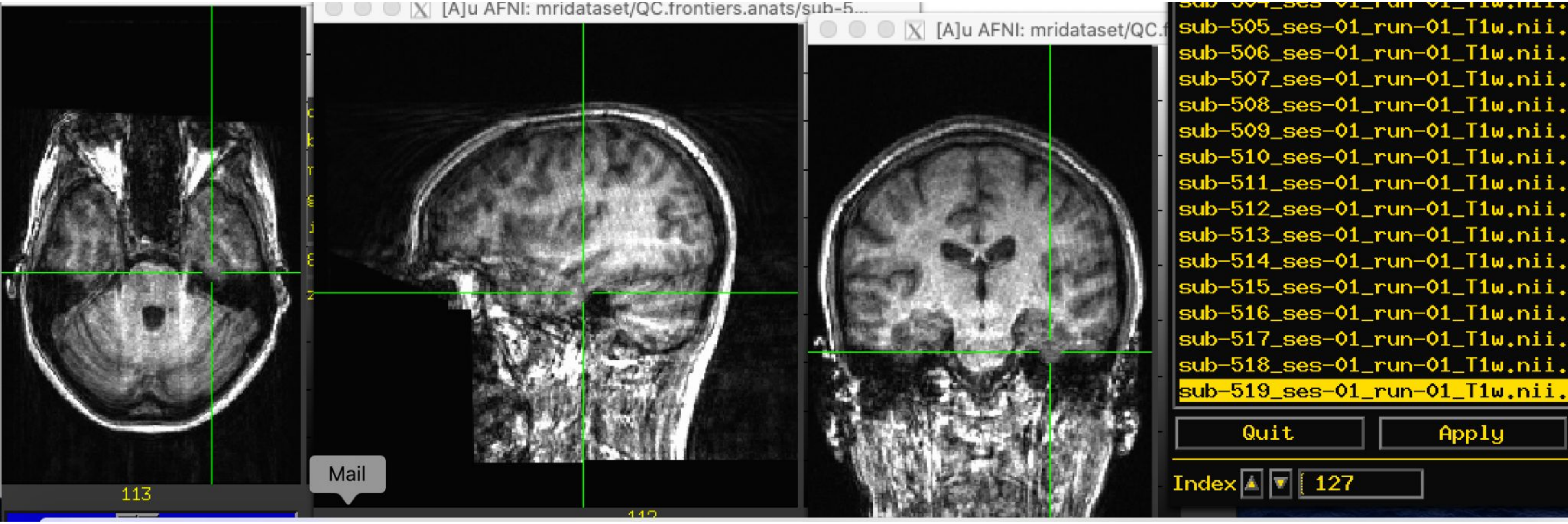
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[A]u AFNI: mridataset/QC.f
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sub-507_ses-01_run-01_T1w.nii.
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sub-518_ses-01_run-01_T1w.nii.
sub-519_ses-01_run-01_T1w.nii.
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